HARSHAL RAAJ PATNAIK

Senior Full-Stack & Cloud Systems Engineer | Technical Product & Platform Lead

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EDUCATION

Indian Institute of Management Kozhikode (IIMK) **2025 – Present**

Executive Post Graduate Programme in Management (EPGPM) – Business Strategy, Platform Economics & Operations *Kochi, India*

SRM Institute of Science and Technology **2017 – 2021**

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.76 / 87.6%) *Chennai, India*

PROFESSIONAL EXPERIENCE

Tata Communications Ltd. **Jul 2021 – Present**

Software Development Engineer (Full-Stack & Cloud Platforms) *Chennai, India*

- Reduced cloud batch scheduler execution latency by **92.6% (slashing cycle time from ~18 minutes down to 80 seconds)** across distributed cloud clusters by partitioning Cassandra mutations into strict 50-item batches, implementing multi-level Redis caching, and tuning HikariCP connection pools.
- Architected the full-stack onboarding workflow for **Caringo Swarm S3-compatible cloud object storage** using Java, Maven, and Angular, automating bucket provisioning and dynamic DNS A-record associations for **100% self-service cloud storage rollout**.
- Engineered multi-hypervisor orchestration dashboard managing heterogeneous **KVM, VMware ESXi, and OLVM clusters**; decoupled long-running provisioning tasks via asynchronous **RabbitMQ message queues**, ensuring zero UI blocking and reliable state reconciliation.
- Owned and scaled the core enterprise VM tracking service serving **executive leadership**, delivering real-time virtual infrastructure capacity tracking, state reconciliation, and multi-tenant host allocation.
- Automated perimeter security provisioning via **FortiGate REST APIs** for firewall rules and IPsec VPNs; engineered an in-process sliding-window rate limiter that **eliminated 100% of HTTP 429 rate-limiting failures**.
- Spearheaded frontend modernization from legacy AngularJS to modern **Angular 17**, re-architecting monolithic views into reactive, modular components and **reducing client-side render times and maintenance overhead**.
- Automated continuous integration and artifact deployment pipelines via **Jenkins CI**, reducing release cycle friction and enforcing automated **JaCoCo code quality and regression gates**.
- Stepped up as **Acting Scrum Master** facilitating sprint ceremonies, cross-functional dependency tracking, and timely production delivery; earned **3× Extra Mile Commitment Awards** for operational excellence.

STRATEGY & TECHNICAL PRODUCT CAPSTONE

SthiraX: Distributed CO_2-Mineralization Platform for Certified Precast Materials | *IIM Kozhikode BPP* **In Progress**

- Architected an asset-light climate-tech platform retrofitting regional precast plants with modular industrial CO_2-curing units, targeting India's **mandatory 5%–25% recycled precast mandate by 2030 across 20,000+ sqm projects** (addressing India's <1% recycling baseline).
- Engineered batch-level **MRV (Measurement, Reporting & Verification) software** generating cryptographically verifiable Digital Product Passports for developers and EPC contractors fulfilling legal EPR compliance.
- Designed an asset-light multi-tiered monetization strategy (equipment leasing, per-tonne curing royalties, and MRV SaaS subscriptions), modeling viable unit economics independent of voluntary carbon credit market volatility.
- Developed a geospatial multi-factor ranking algorithm evaluating industrial CO_2 haul economics, steel slag feedstock supply, and regional precast manufacturing cluster density.

TECHNICAL & LEADERSHIP COMPETENCIES

Core Languages: Java (8/11/17), Python, TypeScript, JavaScript, SQL
Backend & Distributed Systems: Spring Boot 3, Hibernate/JPA, RabbitMQ (AMQP), REST APIs, Microservices Architecture, Flowable BPMN
Frontend & UI Engineering: Angular 17, AngularJS, RxJS, HTML5/CSS3, Tailwind CSS, Responsive Web Design, State Management
Databases, Caching & Cloud: MySQL, Cassandra, Redis, Caringo Swarm / S3, KVM, VMware ESXi, OLVM, Docker
DevOps, SRE & Security: Jenkins CI/CD, Git, Linux, Maven, FortiGate APIs, Postman, Swagger/OpenAPI, SonarQube, JaCoCo
Product Strategy & Leadership: Platform Business Models, Unit Economics, EPR Compliance, Agile / Scrum Master, Cross-Functional Leadership

HONORS, CERTIFICATIONS & EXTRACURRICULAR

- **3× Extra Mile Commitment Award** – Tata Communications Ltd. for critical cloud platform delivery and reliability.
- **Flipkart Grid 2.0 Engineering Challenge** – Team Lead; achieved **Top 150 Nationwide Rank** out of 20,000+ teams.
- **Architecting with Google Compute Engine** (Google Cloud / Coursera) & **Full Stack Web Development** (Coding Ninjas).
- **Rank 2404 Globally** in Second Round of 5th SOF International Mathematics Olympiad (Science Olympiad Foundation).